📉 WhyDrop

The agent that can explain *why* sales dropped, and proves every claim.

GenAI Works Open Agent Hackathon 2026 · Track 2 · Nathan Melia · github.com/jamesbrowndevs/whydrop

Problem & users

- Small online sellers (Gumroad, Shopify, Etsy) watch revenue fall with no analyst to ask.
- Dashboards show **what** dropped, not **why**.
- The clues are scattered: order exports, ad spend, traffic, refunds DB, price and release logs.
- Generic chatbots guess, and an unverified "why" leads to wrong fixes.

Solution

Ask in plain English: *"Why did revenue drop in September?"*

- Ranked causes with confidence and estimated $ impact
- Every claim cites exact cells (query · row · column · value)
- Rejected hypotheses shown with their evidence
- A recommended action per cause; Markdown report export

How it works

① Signal scan on Zetaris → ② Plan hypotheses + recall memory → ③ SQL per hypothesis

④ Analyse → cited claims → ⑤ Verify every claim → ⑥ Ranked report → ⑦ Remember (H-MEM)

Hypotheses are tested in parallel; the UI streams every step, query and model call live.

Architecture

Layer	Tech	Role
Data	Zetaris Lightning (Spark)	WHYDROP_FILES (CSV/JSON filestore) + WHYDROP_PG (Postgres JDBC) federated; the agent's only tool is zetaris_sql()
Reasoning	NVIDIA Nemotron	planner & analyst: 3 Super 120B · SQL: 3 Nano reasoning · verifier: 3.5 Lightning, with routing + fallback
Memory	Meterless H-MEM	verified causes and rejections stored with provenance and recalled next run
UI	Streamlit	live steps, SQL + rows, routing, ranked report

Trust: no claim without evidence

1. The cited query is re-run on Zetaris, and the result hash must reproduce
2. Each cited cell must exist and match
3. Every number in the text must come from cited cells (or a simple delta, ratio or %)
4. An independent Nemotron verifier lane must agree

Failing claims are struck out and never reach the answer.

Sponsor tech

- **Zetaris**: one SQL layer over files + Postgres; cross-source joins; verification re-runs
- **Meterless**: H-MEM long-term memory of investigations (agentathon starter engine)
- **NVIDIA Nemotron**: four lanes, routing visible in UI, logs and reports
- **Cursor**: development environment

Results

Synthetic seller dataset (Jun–Sep 2026) with 3 planted causes + 2 red herrings; the agent never sees the ground truth. Only verified claims are scored.

Question	Result	Causes	Rejected	Time
Why did revenue drop in September 2026?	✅ pass	4	3	542s
Is our ad spend still working?	✅ pass	3	2	528s
Why did Notion Freelance OS sales fall even though we raised the price?	✅ pass	4	3	807s
Why are refunds up in the second half of September?	✅ pass	1	5	842s
Did email marketing cause the drop?	✅ pass	5	2	778s

5/5 passed · 3/3 planted causes found on the headline question · run 2026-10-24 02:08 BST

What's unique

- Explains *and* rejects: shows the hypotheses it ruled out, with proof
- Deterministic + LLM verification of every number
- Federated data without ETL, via Zetaris
- Learns across investigations with Meterless memory

Future work

- Live Shopify / Gumroad / Stripe connectors
- Watchdog mode: alert on a drop with the explanation attached
- Causal-impact estimates; hosted multi-tenant version

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